

classical computers, including supercomputers, could only be used to further quantum technologies to a certain degree because no classical system could be used to comprehend the behaviour of even slightly complex quantum systems.

The cutting-edge field of quantum computing uses the ideas of quantum mechanics to process data in ways that are essentially distinct from those of traditional, or classical, computing. Quantum computers use quantum bits, or qubits, which can exist in multiple states simultaneously, in contrast to classical computers that process information in binary form, representing data as either 0s or 1s. This special property creates new computational opportunities by enabling quantum computers to complete some tasks tenfold faster than classical systems. Cryptography, material science, artificial intelligence, and complex optimisation are just a few of the domains that will be greatly impacted by quantum computing.

Key concepts in quantum computing

The basic element of quantum computing is qubits. Similar to bits in classical computing, qubits are the basic building blocks of information in quantum computing. Qubits can exist in a simultaneous superposition of both states, unlike classical bits, which are only able to exist in one of two states: 0 or 1. Because of this characteristic, quantum computers can process and store vast amounts of data with fewer qubits than classical computers. A variety of physical systems, including photons, superconducting circuits, and trapped ions, can be used to create qubits; each has pros and cons of its own.

Superposition: One of the main ideas that sets quantum computing apart from classical computing is superposition, which is a fundamental principle of quantum mechanics. A qubit can simultaneously exist in

a combination of the states 0 and 1 in superposition. This is expressed mathematically as a linear combination of the two states, which enables qubits to represent several values at once. Quantum computers can execute parallel computations thanks to superposition, which raises the potential speed at which complex problems can be solved.

Entanglement: Another unusual quantum phenomenon is entanglement, which occurs when two or more qubits become correlated to the point where, independent of their distance from one another, the states of the two qubits are directly related. Entangled qubits form a strong bond that can be used in quantum algorithms because measuring the state of one qubit instantly determines the state of the other. Entanglement is necessary for some quantum algorithms and cryptographic protocols, and enables more intricate operations.

Quantum gates: ‘Quantum gates’, which are comparable to logic gates in classical computing, are used in quantum computing to carry out operations on qubits. Quantum gates frequently use rotations and transformations to change the probabilities of qubit states. The Pauli-X gate, which reverses a qubit’s state; the Hadamard gate, which produces superposition; and the CNOT (Controlled NOT) gate, which entangles two qubits are a few examples. Quantum circuits are created by applying sequences of gates to qubits to perform calculations; quantum gates are commonly represented as matrices.

Quantum algorithms: To solve problems more quickly than classical algorithms, ‘quantum algorithms’ are specifically made to take advantage of the special qualities of qubits, such as superposition and entanglement. Notable quantum algorithms that exhibit significant speedups over their classical counterparts are Grover’s algorithm for searching unsorted databases and Shor’s algorithm for factoring large numbers. Exploring the

possibilities of quantum computing in fields where traditional methods fall short is based on quantum algorithms.

Quantum systems: Since ‘quantum systems’ are highly sensitive to their surroundings, interactions can result in decoherence, or the loss of a qubit’s quantum state. Due to the rapid degradation of the delicate states needed for precise computation, quantum computations are susceptible to errors. Decoherence is being countered by quantum error correction techniques, which will enable qubits to carry out more intricate calculations and retain their state for longer. In order to construct scalable and dependable quantum computers, error correction is essential.

Quantum speedup: The potential for quantum computers to solve specific kinds of problems far more quickly than traditional computers is known as ‘quantum speedup’. In domains like materials science and cryptography, where conventional computing techniques might not be adequate to address the scope of the issues, quantum speedup holds promise.

Quantum hardware: Specialised hardware that can produce and manipulate qubits while reducing external interference is needed to build a working quantum computer. Topological qubits, trapped ions, and superconducting circuits are some of the various methods used to create qubits. Coherence time and scalability are two examples of the advantages and difficulties that are specific to each technology. To make quantum computing a viable technology, developments in quantum hardware are essential.

The key differences between classical programming and quantum programming are listed in Table 1.

Quantum computing can solve problems that even the most powerful classical computers cannot. Key concepts such as qubits, superposition, entanglement, quantum gates, quantum algorithms, quantum systems, quantum